

L13



L13: Gauss' Lemma continued... ①

Proof (of Gauss' Lemma): Let $r_1, r_2, \dots, r_n$ be the least non-negative residues of the integers $a, 2a, \dots, \left(\frac{p-1}{2}\right)a$ that are greater than $p/2$ and let $s_1, s_2, \dots, s_m$ be the least non-negative residues of these integers that are less than $p/2$ (no least positive residue is equal to $p/2$). Note that no r_i or s_j is equal to zero, since p does not divide any $a, 2a, \dots, \left(\frac{p-1}{2}\right)a$. Consider the $\frac{p-1}{2}$ integers given by

$$p - r_1, p - r_2, \dots, p - r_n, s_1, s_2, \dots, s_m. \quad \text{①}$$

We wish to show that these integers are all the integers from 1 to $\frac{p-1}{2}$ inclusive (in some order).

Since each integer in ① is between 1 and $\frac{p-1}{2}$ - inclusive, it suffices to prove

that no two of these integers are congruent modulo p . No two of the first n integers are congruent modulo p , otherwise $p - r_i \equiv p - r_j \pmod{p}$ for some $i \neq j$, and then $r_i \equiv r_j \pmod{p}$. Thus $k_i a \equiv k_j a \pmod{p}$ for some $k_i, k_j \in \mathbb{Z}$ with $k_i \neq k_j$ where $1 \leq k_i \leq \frac{p-1}{2}$ and $1 \leq k_j \leq \frac{p-1}{2}$. Since $p \nmid a$, the inverse of a modulo p exists, and thus $k_i \equiv k_j \pmod{p}$. Contradiction! Similarly, no two of the second m integers are congruent modulo p . Finally, no one of the first n integers can be congruent modulo p to one of the second m integers, otherwise $p - r_i \equiv s_j \pmod{p}$ for some i and j , and so $-k_i a \equiv k_j a \pmod{p}$ for some $k_i, k_j \in \mathbb{Z}$ with $k_i \neq k_j$ and $1 \leq k_i, k_j \leq \frac{p-1}{2}$. Applying the inverse of a

(2)

modulo p gives $-k_i \equiv k_j \pmod{p}$. (3)

Contradiction!

Thus the integers in (1) are the integers from 1 to $\frac{p-1}{2}$ inclusive (in some order). Then

$$(p-r_1)(p-r_2)\cdots(p-r_n) s_1 s_2 \cdots s_m \equiv \left(\frac{p-1}{2}\right)! \pmod{p}.$$

Thus

$$(-1)^n r_1 r_2 \cdots r_n s_1 s_2 \cdots s_m \equiv \left(\frac{p-1}{2}\right)! \pmod{p}.$$

By definition of the r_i and s_j , we have

that

$$(-1)^n (a)(2a)\cdots\left(\frac{p-1}{2}a\right) \equiv \left(\frac{p-1}{2}\right)! \pmod{p},$$

or equivalently,

$$(-1)^n a^{(p-1)/2} \left(\frac{p-1}{2}\right)! \equiv \left(\frac{p-1}{2}\right)! \pmod{p}$$

Since $((\frac{p-1}{2})!, p) = 1$, the inverse of $(\frac{p-1}{2})!$

modulo p exists, and thus we obtain

$$(-1)^n a^{(p-1)/2} \equiv 1 \pmod{p},$$

and so $a^{(p-1)/2} \equiv (-1)^n \pmod{p}$. (2)

Euler's criterion and (2) imply that (4)
 $\left(\frac{a}{p}\right) \equiv (-1)^n \pmod{p}$. Since $\left(\frac{a}{p}\right) = \pm 1$, we
 have that $\left(\frac{a}{p}\right) = (-1)^n$, as desired. $\square$

Theorem 4.8: Let p be an odd prime number.

Then $\left(\frac{2}{p}\right) = (-1)^{(p^2-1)/8} = \begin{cases} 1 & \text{if } p \equiv 1, 7 \pmod{8} \\ -1 & \text{if } p \equiv 3, 5 \pmod{8} \end{cases}$.

Proof: By Gauss Lemma, we have that

$\left(\frac{2}{p}\right) = (-1)^n$ where n is the number of least
 positive residues of the integers

$$2, (2)2, \dots, \left(\frac{p-1}{2}\right)2 \quad (3)$$

that are greater than $p/2$. Let $k \in \mathbb{Z}$ with
 $1 \leq k \leq \frac{p-1}{2}$. Then $k(2) < p/2$ if and only if

$k < p/4$. Thus only $\left[\frac{p}{4}\right]$ integers from (3)

are less than $p/2$. Thus $\frac{p-1}{2} - \left[\frac{p}{4}\right]$ integers

in (3) are greater than $p/2$, from which

$$\left(\frac{2}{p}\right) = (-1)^{\frac{p-1}{2} - \left[\frac{p}{4}\right]} \text{ by Gauss' Lemma. } \textcircled{5}$$

It Suffices to prove that

$$\frac{p-1}{2} - \left[\frac{p}{4}\right] \equiv \frac{p^2-1}{8} \pmod{2}. \textcircled{4}$$

Case 1: $p \equiv 1 \pmod{8}$: Then $p = 8k+1$ for some $k \in \mathbb{Z}$, from which

$$\begin{aligned} \frac{p-1}{2} - \left[\frac{p}{4}\right] &= \frac{8k+1-1}{2} - \left[\frac{8k+1}{4}\right] \\ &= 4k - 2k = 2k \equiv 0 \pmod{2}. \end{aligned}$$

On the other hand,

$$\frac{p^2-1}{8} = \frac{(8k+1)^2-1}{8} = 8k^2 + 2k \equiv 0 \pmod{2}.$$

Thus $\textcircled{4}$ holds when $p \equiv 1 \pmod{8}$.

Establishing $\textcircled{4}$ in the cases $p \equiv 3, 5, 7 \pmod{8}$ follow by a similar computation and are left as an exercise. This establishes the first equality in the Theorem. The second equality

in the Theorem follows from the first. (6)

Example: We have $\left(\frac{2}{7}\right) = 1$ by Theorem 4.8, since $7 \equiv 7 \pmod{8}$.

• We have $\left(\frac{2}{53}\right) = -1$ by Theorem 4.8, since $53 \equiv 5 \pmod{8}$.

Recap: • We know how to compute $\left(\frac{-1}{p}\right)$ and $\left(\frac{2}{p}\right)$.

• We now need to understand $\left(\frac{q}{p}\right)$ for p, q both odd primes and $p \neq q$.

Quadratic Reciprocity

• The law of quadratic reciprocity relates $\left(\frac{q}{p}\right)$ to $\left(\frac{p}{q}\right)$, which sometimes simplifies computations!

Theorem 4.9 (Law of Quadratic reciprocity)

Let p and q be distinct odd prime numbers.

Then

(7)

$$\begin{aligned} \left(\frac{p}{q}\right)\left(\frac{q}{p}\right) &= (-1)^{\left(\frac{p-1}{2}\right)\left(\frac{q-1}{2}\right)} \\ &= \begin{cases} 1 & \text{if } p \equiv 1 \pmod{4} \text{ or } q \equiv 1 \pmod{4} \\ & \text{(or both)} \\ -1 & \text{if } p \equiv q \equiv 3 \pmod{4} \end{cases} \end{aligned}$$

Remark: If $p \equiv 1 \pmod{4}$ or $q \equiv 1 \pmod{4}$, then $\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = 1$ and so $\left(\frac{p}{q}\right) = \left(\frac{q}{p}\right)$.

If $p \equiv q \equiv 3 \pmod{4}$, then $\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = -1$, so that $\left(\frac{p}{q}\right) = -\left(\frac{q}{p}\right)$.

Example: Compute $\left(\frac{7}{53}\right)$. We have that

$$\begin{aligned} \left(\frac{7}{53}\right) &= \left(\frac{53}{7}\right) \quad (\text{by T4.9 and } 53 \equiv 1 \pmod{4}) \\ &= \left(\frac{4}{7}\right) \\ &= 1 \quad (\text{since } 4 \equiv 2^2 \pmod{7}). \end{aligned}$$

(8)

Example: Compute $\left(\frac{-158}{101}\right)$.

Exercise:

Justify the steps.

$$\left(\frac{-158}{101}\right) = \left(\frac{-1}{101}\right) \left(\frac{158}{101}\right)$$

$$= \left(\frac{158}{101}\right)$$

$$= \left(\frac{57}{101}\right)$$

$$= \left(\frac{3}{101}\right) \left(\frac{19}{101}\right)$$

$$= \left(\frac{101}{3}\right) \left(\frac{101}{19}\right)$$

$$= \left(\frac{2}{3}\right) \left(\frac{6}{19}\right)$$

$$= (-1) \left(\frac{6}{19}\right)$$

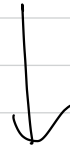
$$= - \left(\frac{2}{19}\right) \left(\frac{3}{19}\right)$$

$$= -(-1) \left(\frac{3}{19}\right)$$

$$= \left(\frac{3}{19}\right)$$

$$= - \left(\frac{19}{3}\right)$$

$$= - \left(\frac{1}{3}\right) = -1.$$



Warning: It would not be correct (9)
to write $\left(\frac{57}{101}\right) = \left(\frac{101}{57}\right)$ by Theorem 4.9.
Why? 57 is not a prime number.